Industrial Electrification Economics

When does switching from gas to electricity become economically rational?

Independent research project and working analytical model

Ruqaiya Motagam

MSc International Energy Studies and Energy Economics | BA Economics & Finance

Executive summary

I developed this project to test the business case for electrifying industrial process heat under current European energy-price conditions and alternative carbon prices. The model compares a natural-gas boiler, an electric boiler and an industrial heat pump for a stylised 1 MWth heat duty operating 4,000 full-load hours per year. It calculates energy demand, energy expenditure, EU ETS carbon cost, capital expenditure, grid-connection cost, annualised capital cost, operations and maintenance, levelised cost of heat (LCOH), payback, net present value (NPV) and operational emissions.

The model uses published 2025-S2 EU non-household energy prices of EUR183.7/MWh for electricity and EUR60.5/MWh for natural gas, an electricity-to-gas price ratio of about 3.04. It uses IEA technology assumptions of 87% gas-boiler efficiency, 98% electric-boiler efficiency and a heat-pump COP of 3.5. Capital and grid costs are editable screening assumptions because they must ultimately be replaced by site-specific vendor and grid-connection quotations.

> *Central screening result: at an illustrative EUR80/tCO2 carbon price, the model produces an LCOH of about EUR93.62/MWh for the gas boiler, EUR199.75/MWh for the electric boiler and EUR86.36/MWh for the industrial heat pump. The heat-pump case has a simple payback of about 6.5 years versus a new gas boiler and a positive 15-year NPV of about EUR249,000 under the stated assumptions.*

The model also demonstrates why electrification cannot be treated as one technology. Electric resistance boilers remain highly exposed to electricity prices, while heat pumps can overcome a high electricity-to-gas price ratio through their much higher efficiency. Process temperature, source-heat availability, grid connection, load factor and capital cost therefore determine whether electrification is practical.

Contents

1. Research question and objectives 4

2. Current policy and market context 4

3. Case definition and technology scope 5

4. Model equations 6

5. Inputs and data sources 7

6. Worked central calculation 8

6.1 Natural-gas boiler 8

6.2 Industrial heat pump 8

6.3 Electric boiler 8

7. Central results 9

8. Carbon-price sensitivity 10

9. Electricity-price and COP sensitivity 10

10. Practical engineering and commercial feasibility 11

11. Site data request for an investment-grade version 12

12. Commercial implementation options 12

13. Carbon accounting and policy treatment 13

14. Key risks and limitations 13

15. Relevance to The Climate Desk 14

16. Accompanying working model 14

References 15

1. Research question and objectives

At what combination of electricity price, natural-gas price, carbon price, process efficiency, heat-pump COP and investment cost does industrial electrification become economically preferable to fossil process heat?

The project has four objectives:

- Quantify the operating-cost effect of the electricity-to-gas price ratio.
- Show how the EU ETS carbon price changes the gas-boiler cost position.
- Separate high-efficiency heat-pump electrification from resistance-based electric boilers.
- Translate the technical comparison into an investment decision using LCOH, payback, NPV and break-even carbon price.

2. Current policy and market context

Industrial electrification has become a central element of EU competitiveness and decarbonisation policy. The European Commission presented its Electrification Action Plan on 17 July 2026. The Commission states that the plan provides a pathway to reduce the electricity-to-gas price gap by 2030 so that electricity prices are not more than twice gas prices for industry. The broader policy package also addresses network charges, taxation, grid deployment, flexibility and smart metering.

The current price gap is material. Eurostat reported an EU average non-household electricity price of EUR18.37 per 100 kWh in the second half of 2025, equivalent to EUR183.7/MWh. The corresponding non-household gas price was EUR6.05 per 100 kWh, equivalent to EUR60.5/MWh. This gives an electricity-to-gas ratio of approximately 3.04 before accounting for conversion efficiency.

The efficiency adjustment is crucial. IEA analysis of industrial heat uses 87% efficiency for a gas boiler, 98% for an electric boiler and a heat-pump COP of 3.5. IEA also reported in 2026 that industrial heat pumps already have lower lifetime costs than gas boilers in 17 EU countries representing around 40% of low-temperature industrial heat demand, with another 35% of demand in countries where the cost difference is within 5%.

The European Commission's 2026 industrial electrification study identifies the remaining barriers as high electricity costs, grid constraints, investment needs, competitiveness concerns and policy uncertainty. My model therefore includes not only energy and carbon costs but also editable grid and capital-cost assumptions.

3. Case definition and technology scope

Parameter	Central case	Why selected
Thermal capacity	1,000 kWth	Large enough to represent an industrial process-heat application while remaining transparent
Full-load hours	4,000 h/year	Moderate industrial utilisation
Useful heat demand	4,000 MWh/year	1 MWth x 4,000 h
Gas boiler	87% efficiency	IEA industrial heat assumption
Electric boiler	98% efficiency	IEA industrial heat assumption
Industrial heat pump	COP 3.5	IEA central industrial heat assumption
Project life	15 years	Common screening horizon
WACC	8%	Lower end of IEA 8%-12% range used in industrial heat analysis

The heat-pump case is most applicable to low- and medium-temperature process heat where a suitable source of waste or ambient heat is available. IEA and JRC evidence indicates that commercially available industrial heat pumps are particularly relevant below roughly 120-150 degrees C, with higher-temperature applications developing rapidly but requiring more site-specific engineering.

4. Model equations

```
Annual useful heat (MWh) = Thermal capacity (kWth) x Full-load hours / 1,000

Gas input (MWh) = Useful heat / Gas-boiler efficiency

Electric-boiler electricity (MWh) = Useful heat / Electric-boiler efficiency

Heat-pump electricity (MWh) = Useful heat / COP

Direct gas emissions (tCO2) = Gas input x 0.202 tCO2/MWh

Carbon cost = Direct emissions x EU ETS carbon price

Capital recovery factor (CRF) = r(1+r)^n / ((1+r)^n - 1)

Annualised CAPEX = Net CAPEX x CRF

Total annual cost = Energy cost + Carbon cost + Annualised CAPEX + O&M

LCOH = Total annual cost / Useful heat supplied

Simple payback = Incremental upfront CAPEX / Annual operating-cost saving

NPV = -Incremental CAPEX + sum(Annual operating saving / (1+r)^t)

Break-even carbon price = (Electrified LCOH - Gas LCOH excluding carbon) / Gas direct-emissions intensity per MWh useful heat
```

5. Inputs and data sources

Input	Central value	Status
EU non-household electricity price	EUR183.7/MWh	Eurostat 2025-S2
EU non-household natural-gas price	EUR60.5/MWh	Eurostat 2025-S2
Germany electricity benchmark	EUR226.4/MWh	Eurostat 2025-S2
Germany gas benchmark	EUR71.3/MWh	Eurostat 2025-S2
EU ETS carbon price	EUR80/tCO2	Illustrative scenario, not forecast
Gas emission factor	0.202 tCO2/MWh fuel	JRC/IPCC
EU electricity GHG intensity	0.210 tCO2e/MWh	EEA 2024 early estimate
Gas boiler CAPEX	EUR150/kWth	Editable author assumption
Electric boiler CAPEX	EUR200/kWth	Editable author assumption
Heat pump CAPEX	EUR900/kWth	Editable author assumption
Electric-boiler grid cost	EUR200/kWe	Editable author assumption
Heat-pump grid cost	EUR100/kWe	Editable author assumption

I explicitly separate sourced market/technology inputs from author assumptions. The CAPEX, O&M and grid values are not presented as market facts. They are placeholders that allow the economic model to function and must be replaced with project-specific quotations for a real investment appraisal.

6. Worked central calculation

6.1 Natural-gas boiler

```
Useful heat = 1,000 kW x 4,000 h / 1,000 = 4,000 MWh/year
Gas input = 4,000 / 0.87 = 4,597.70 MWh/year
Energy cost = 4,597.70 x EUR60.5 = EUR278,161/year
Direct emissions = 4,597.70 x 0.202 = 928.74 tCO2/year
Carbon cost at EUR80/tCO2 = 928.74 x 80 = EUR74,299/year
Annualised CAPEX (EUR150,000 at 8%, 15y) = approximately EUR17,524/year
O&M = 3% x EUR150,000 = EUR4,500/year
Total annual cost = EUR374,484; LCOH = EUR93.62/MWh useful heat
```

6.2 Industrial heat pump

```
Electricity use = 4,000 / 3.5 = 1,142.86 MWh/year
Electricity cost = 1,142.86 x EUR183.7 = EUR209,943/year
Equipment CAPEX = EUR900/kWth x 1,000 kWth = EUR900,000
Grid CAPEX = EUR100/kWe x (1,000/3.5) = EUR28,571
Annualised net CAPEX = approximately EUR108,485/year
O&M = 3% x EUR900,000 = EUR27,000/year
Total annual cost = EUR345,427; LCOH = EUR86.36/MWh useful heat
```

Using the EU-average 2024 electricity GHG intensity of 0.210 tCO2e/MWh as a screening factor, the heat pump has approximately 240 tCO2e/year of Scope 2 emissions, compared with 928.7 tCO2/year of direct gas emissions in the boiler case. This is a screening comparison rather than a contractual Scope 2 accounting result.

6.3 Electric boiler

```
Electricity use = 4,000 / 0.98 = 4,081.63 MWh/year
Electricity cost = 4,081.63 x EUR183.7 = EUR749,796/year
Central LCOH = EUR199.75/MWh useful heat
```

The electric boiler illustrates why electrification economics depend on conversion technology. Near-100% resistance efficiency cannot offset an electricity-to-gas price ratio above 3, while a heat pump can because each MWh of electricity produces multiple MWh of useful heat.

7. Central results

Technology	LCOH	Annual cost	Operational emissions	Interpretation
Natural-gas boiler	EUR93.62/MWh	EUR374,484	928.74 tCO2 direct	Baseline with EUR80/t carbon cost
Electric boiler	EUR199.75/MWh	EUR799,005	857.14 tCO2e Scope 2	Not competitive at EU average electricity price
Industrial heat pump	EUR86.36/MWh	EUR345,427	240.00 tCO2e Scope 2	Lowest central-case LCOH

For the heat pump relative to a new gas boiler:

- Incremental upfront CAPEX: approximately EUR778,571.
- Annual operating-cost saving at central prices: approximately EUR120,017.
- Simple payback: approximately 6.5 years.
- 15-year NPV at 8%: approximately +EUR248,711.
- Break-even carbon price: approximately EUR48.71/tCO2, holding other assumptions constant.
- Operational emissions reduction using EU-average grid intensity: approximately 74.2%.

8. Carbon-price sensitivity

Carbon price	Gas boiler LCOH	Electric boiler LCOH	Heat-pump LCOH
EUR0/t	EUR75.05	EUR199.75	EUR86.36
EUR50/t	EUR86.66	EUR199.75	EUR86.36
EUR80/t	EUR93.62	EUR199.75	EUR86.36
EUR100/t	EUR98.26	EUR199.75	EUR86.36
EUR150/t	EUR109.87	EUR199.75	EUR86.36
EUR200/t	EUR121.48	EUR199.75	EUR86.36

This table isolates the carbon-price channel. At zero carbon price, the gas boiler remains cheaper on the full central LCOH because the heat pump carries materially higher capital cost. At around EUR49/tCO2, the model reaches cost parity. The electric boiler remains far above both because electricity cost dominates its economics.

9. Electricity-price and COP sensitivity

The heat-pump break-even carbon price changes materially with electricity price and COP. Negative values mean the heat pump is already cheaper than gas before applying a carbon price under the stated capital-cost assumptions.

Electricity price	COP 2.0	COP 2.5	COP 3.0	COP 3.5	COP 4.0
EUR100/MWh	EUR40.7	-EUR3.6	-EUR33.2	-EUR54.3	-EUR70.1
EUR140/MWh	EUR126.8	EUR65.3	EUR24.3	-EUR5.1	-EUR27.0
EUR180/MWh	EUR213.0	EUR134.2	EUR81.7	EUR44.2	EUR16.0
EUR220/MWh	EUR299.1	EUR203.1	EUR139.1	EUR93.4	EUR59.1
EUR260/MWh	EUR385.3	EUR272.0	EUR196.5	EUR142.6	EUR102.2

The sensitivity table shows that COP is not a minor technical parameter; it is an economic driver. A process requiring a low COP because of a large temperature lift can lose the business case even when a strong carbon price exists. Conversely, an application with an accessible waste-heat source and COP above 3.5 can remain attractive at much higher electricity prices.

10. Practical engineering and commercial feasibility

A positive spreadsheet result is not enough to justify electrification. I would test the following practical conditions before treating the heat-pump case as investable:

- Process temperature: confirm the required supply and return temperatures and whether they remain within commercially proven heat-pump ranges.
- Heat-source availability: identify stable waste heat, cooling-water, ambient, wastewater or other source streams and their temperature profile.
- Load profile: confirm hourly and seasonal heat demand. High full-load hours materially improve capital recovery.
- Integration: determine whether the process can accept lower-temperature heat directly or requires steam generation and additional equipment.
- Grid connection: verify available electrical capacity, transformer rating, connection queue, reinforcement cost and lead time.
- Electrical tariff: compare fixed retail, wholesale-indexed, PPA and flexible-operation structures rather than relying only on an average Eurostat price.
- Flexibility and storage: assess thermal storage, hybrid operation and load shifting into low-price electricity hours.
- Downtime and production risk: quantify installation shutdown, commissioning and contingency requirements.
- Refrigerant and safety requirements: assess refrigerant selection, F-gas implications, pressure systems and maintenance capability.
- Vendor bankability: obtain performance guarantees for COP at the actual temperature lift, not catalogue COP at different conditions.

11. Site data request for an investment-grade version

Data required	Unit / frequency	Purpose
Heat demand	Hourly MWhth or steam flow	Size equipment and determine load factor
Supply/return temperatures	Degrees C, hourly where variable	Determine technical COP and technology suitability
Waste-heat source temperature/flow	Degrees C and flow	Heat-pump source feasibility
Gas consumption and invoice	MWh and EUR, monthly	Site-specific baseline cost
Electricity consumption and invoice	MWh, peak kW, tariff	Marginal electricity cost and grid headroom
Grid connection capacity	kVA/MVA	Connection feasibility and reinforcement need
Existing boiler efficiency	Measured or audited %	Correct baseline
Maintenance cost	EUR/year	O&M comparison
Vendor CAPEX quote	EUR by package	Replace screening assumptions
Production outage cost	EUR/hour or EUR/day	Integration and commissioning risk
EU ETS exposure	Verified tCO2 and allowance strategy	Carbon-cost baseline
Financing terms	WACC, hurdle rate, debt terms	Investment appraisal

12. Commercial implementation options

If a full heat-pump conversion is not immediately viable, I would test intermediate strategies:

- Hybrid gas + heat-pump system: use the heat pump for baseload and gas for peaks/high-temperature duty.
- Thermal storage: decouple heat production from instantaneous process demand and exploit lower-price electricity periods.

- PPA or on-site renewables: reduce long-run electricity-price exposure where technically and contractually feasible.
- Flexible electric boiler: operate only in the cheapest electricity hours when a complementary heat source or storage exists.
- Phased electrification: start with the lowest-temperature process streams and expand after grid reinforcement or technology upgrades.
- Efficiency first: reduce heat demand and improve heat recovery before sizing new electrified equipment.

13. Carbon accounting and policy treatment

The gas case bears direct combustion emissions and, where the installation is covered by the EU ETS, these emissions create allowance demand. The electrified cases move emissions from direct combustion to electricity generation. My model therefore reports direct and Scope 2 emissions separately. The central Scope 2 comparison uses the EEA EU-average 2024 electricity GHG intensity of 210 gCO2e/kWh as a screening value.

A real corporate decision would require the appropriate location-based and market-based Scope 2 methods, contractual electricity sourcing, future grid decarbonisation and any interaction with product carbon-footprint or CBAM methodologies. The screening model should therefore not be used as a final product-carbon-footprint calculation.

14. Key risks and limitations

- Eurostat energy prices are benchmark averages; a real industrial site may pay materially different commodity, network and tax components.
- The CAPEX, O&M and grid-connection values are author assumptions and must be replaced with quotes.
- COP varies with source and sink temperature, part load, ambient conditions and equipment design.
- The model assumes constant annual prices rather than hourly dispatch or stochastic price paths.
- Grid-emissions intensity is an average screening metric, not a marginal emissions factor.
- No production-value loss is assigned to installation downtime.
- No residual value, depreciation, tax shield or financing structure is included.
- Carbon-price scenarios are illustrative and not a forecast of EU allowance prices.

15. Relevance to The Climate Desk

This project demonstrates the link between carbon pricing, energy prices and real industrial investment decisions. That is directly relevant to The Climate Desk's work on EU ETS reform, industrial decarbonisation and the economic conditions needed to unlock clean investment. The model also shows how I would communicate a policy question to a client: start with a transparent cost model, separate sourced inputs from assumptions, run sensitivities, identify implementation barriers and translate the results into decision criteria.

16. Accompanying working model

The Excel workbook contains:

- README: scope, decision question and boundaries.
- Inputs: public market/technology values and editable project assumptions.
- Technology Model: energy, carbon, CAPEX, annual cost, LCOH and emissions.
- Investment Case: incremental CAPEX, operating savings, payback, NPV and break-even carbon price.
- Sensitivity: electricity-price/COP and carbon-price scenarios.
- Dashboard: central KPIs and charts.
- Sources: direct URLs and modelling notes.

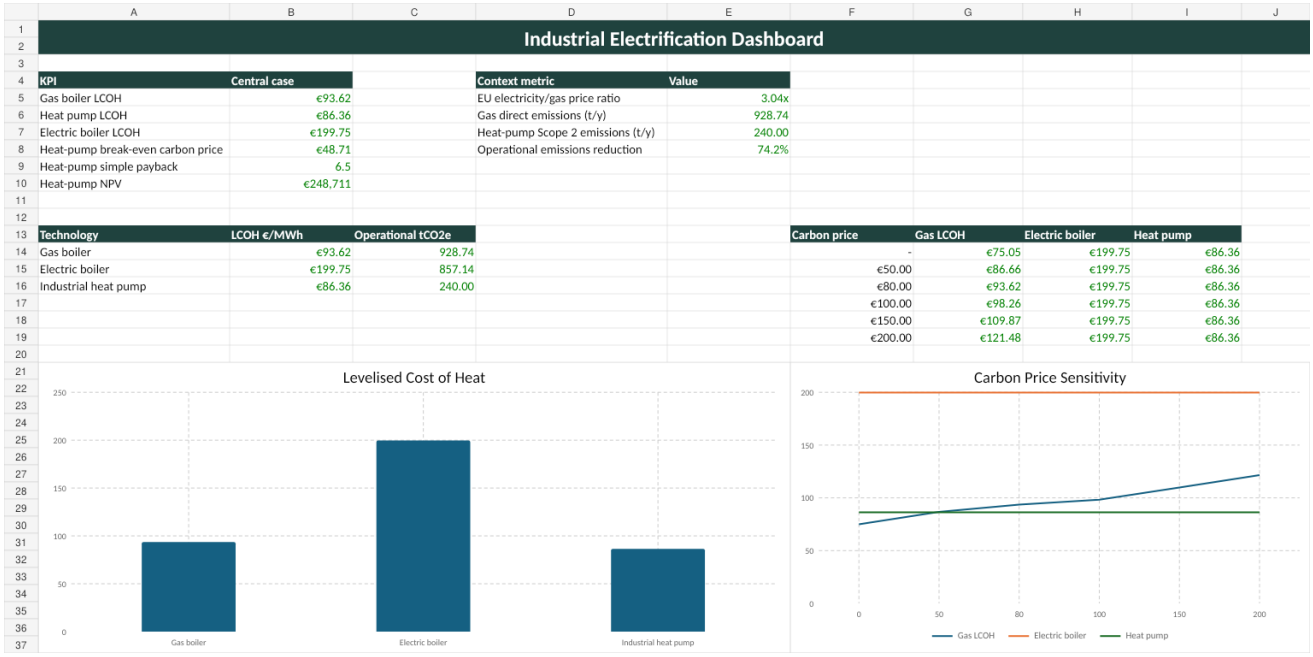


Figure 1. Industrial electrification model dashboard - central case and carbon-price sensitivity.

References

[1] Eurostat, non-household electricity prices in second half of 2025: https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20260508-2

[2] Eurostat, non-household gas prices in second half of 2025: https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20260508-3

[3] European Commission, Electricity prices and 2026 Electrification Action Plan: https://energy.ec.europa.eu/topics/markets-and-consumers/electricity-prices_en

[4] European Commission, Study on pathways and policies for accelerating electrification and use of renewables in industry, 21 July 2026: https://energy.ec.europa.eu/publications/study-pathways-and-policies-accelerating-electrification-and-use-renewables-industry_en

[5] IEA, Can low-temperature heat in factories be electrified competitively?: https://www.iea.org/commentaries/can-low-temperature-heat-in-factories-be-electrified-competitively

[6] IEA, The energy crisis creates even stronger impetus for EU electrification, 2026: https://www.iea.org/commentaries/the-energy-crisis-creates-even-stronger-impetus-for-eu-electrification

[7] European Commission JRC, default fuel emission factors: https://publications.jrc.ec.europa.eu/repository/bitstream/JRC57789/com%20guidebook%20jrc%20format.pdf

[8] European Environment Agency, EU electricity GHG intensity, 2024 early estimate: https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emission-intensity-of-1-1751032678/greenhouse-gas-emission-intensity-of-electricity-generation-eu-level